



The impact of local government debt on corporate inefficient investment

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ABSTRACT

In recent years, the rapid expansion of local government debt and the accumulation of debt risks have posed potential threats to China's long-term economic stability. Using a matched dataset of China's A-share listed firms and prefecture-level city macroeconomic data from 2008 to 2021, this paper systematically investigates the impact of local government debt on corporate inefficient investment and its underlying mechanisms. The empirical results show that the expansion of local government debt significantly increases corporate inefficient investment, primarily by exacerbating corporate over-investment. Mechanism analyses indicate that: (1) local government debt expansion stimulates public investment and amplifies corporate inefficient investment through the fiscal investment multiplier effect; (2) local government debt increases corporate financialization, which partially mitigates inefficient investment; and (3) local government debt reduces firms' tax burdens, thereby reinforcing corporate inefficient investment. Further heterogeneity analyses reveal that the adverse effect of local government debt on inefficient investment is more pronounced among firms facing lower financing constraints, weaker internal controls, and larger firm size, as well as in regions with lower fiscal pressure and higher levels of economic development. This study enriches the literature on the microeconomic consequences of local government debt from the perspective of corporate investment efficiency and provides both theoretical and empirical evidence supporting the central government's efforts to strengthen local government debt regulation and prevent systemic risks.

1. Introduction

Since the implementation of China's tax-sharing reform, local governments have faced pronounced fiscal imbalances under an institutional framework characterized by centralized revenue authority and decentralized expenditure responsibilities. As a result, debt financing has gradually become an important instrument for local governments to alleviate funding shortages and promote economic growth. In particular, following the 2008 global financial crisis, China implemented countercyclical macroeconomic policies represented by the “four-trillion-yuan” stimulus plan, which effectively mitigated the economic downturn. However, these policies also accelerated the expansion of local government financing vehicles, leading to a substantial increase in local government debt.

As local government debt continues to accumulate, its potential risks have become increasingly salient. Beyond posing challenges to fiscal sustainability, rising local government debt may exert profound effects on long-term regional economic stability. In recent

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years, debt problems in several regions have come to the forefront, intensifying academic and policy debates on the economic consequences of local government debt. Against this background, a systematic evaluation of the economic effects of local government debt and its transmission mechanisms is of great practical importance for preventing debt risks, avoiding resource misallocation, and promoting high-quality economic development.

From a theoretical perspective, the expansion of local government debt may exert dual effects on corporate investment behavior. On the one hand, debt-financed public spending on infrastructure and public services can improve the investment environment and raise the returns to private capital, thereby generating a “crowding-in effect” that stimulates corporate investment (Ciaffi et al., 2024; Kantorowicz et al., 2024; Qu et al., 2023; Yang et al., 2023). Especially during economic downturns or recessions, public investment supported by government debt can stabilize economic activity and promote recovery by expanding aggregate demand (Panizza & Presbitero, 2014; Chen et al., 2026; Cheng et al., 2022). On the other hand, local government debt often carries implicit guarantees, granting local governments a financing advantage over private entities in credit markets (Liu et al., 2023; Tang, 2022; Yang et al., 2023; Zhang, Pan, & Fan, 2022). Excessive borrowing by local governments may crowd out limited financial resources, raise firms' financing costs, and suppress corporate investment (Liu et al., 2023). Moreover, local governments' direct participation in certain markets may further crowd out private investment, exerting adverse effects on long-term economic growth (Fan et al., 2022; Zhang, Pan, & Fan, 2022; Liu et al., 2023).

China has officially entered a stage of high-quality development, in which improving resource allocation efficiency is a core objective. As key microeconomic agents, firms' investment decisions—specifically whether investment deviates from the optimal level—directly affect resource allocation efficiency and economic development quality. Inefficient investment, which manifests as either overinvestment or underinvestment, represents a critical form of resource misallocation at the firm level. Although existing studies have examined the microeconomic effects of local government debt from perspectives such as investment scale, financing structure, and productivity, most of the literature focuses on “how much firms invest” rather than “how efficiently firms invest.” In other words, how local government debt affects corporate inefficient investment remains insufficiently explored. Related studies by Xiong and Shen (2019a, 2019b) and Ma and Zhang (2022) suggest that local government debt exacerbates underinvestment among non-state-owned enterprises and overinvestment among state-owned enterprises, respectively, through distinct channels.

Building on this background and addressing the gaps in the literature, this paper employs prefecture-level local government debt data matched with firm-level data of Chinese A-share listed companies from 2008 to 2021. The results indicate that the expansion of local government debt significantly exacerbates corporate inefficient investment. Mechanism analyses further reveal that government debt results in corporate inefficient investment through fiscal investment multiplier effect, financialization effect and tax reduction effect. In addition, heterogeneity analyses show that the amplifying effect of local government debt on inefficient investment is more pronounced among firms with lower financing constraints, weaker internal controls, and larger size, as well as in regions with lower fiscal pressure and higher economic development levels.

Through the above research, this paper makes contributions to the existing literature in three aspects. First, this paper explores the impact of government debt on corporate inefficient investment. Compared with existing studies that mostly focus on the impact of government debt on corporate investment scale, financing constraints, corporate risk, and corporate innovation (Chen et al., 2024; Fan et al., 2022; Zhang, Pan, & Fan, 2022; Zhou et al., 2023; Zhu et al., 2024), this paper focuses on its impact on corporate inefficiency, expands the relevant research on the impact of public debt, and provides policy implications for improving corporate investment efficiency. Second, this paper reveals the mechanism of government debt affecting corporate inefficient investment from multiple dimensions and responds to relevant classical theories. Based on classical theories such as the fiscal expenditure multiplier effect, the investment crowding-out effect of public debt, and Ricardian equivalence theorem (Chen et al., 2022; Zheng et al., 2025; Zhu et al., 2022), this paper analyzes the impact mechanism from three dimensions: stimulating corporate investment, accelerating financialization, and reducing corporate tax burden. Most existing studies have revealed the impact mechanism of public debt from the perspective of financing constraints and resource allocation efficiency. Finally, this paper also reveals the heterogeneous impact of government debt on corporate inefficient investment from multiple dimensions. This paper tests the heterogeneity of the effect of public debt among enterprises with different ownership natures, governance levels, and asset sizes, which is conducive to fully revealing this impact.

The remainder of this paper is organized as follows. Section 2 reviews the related literature. Section 3 presents the theoretical analysis and research hypotheses. Section 4 describes the research design, including data sources, variable construction, and model specification. Section 5 reports the empirical results and robustness checks. Section 6 provides mechanism tests and heterogeneity analyses. The final section concludes and discusses policy implications.

2. Literature review

2.1. The effect of local government debt

Existing studies on local government debt mainly focus on two aspects: its impact on macroeconomic growth and its microeconomic effects.

2.1.1. Local government debt and macroeconomic growth

The literature has not reached a consensus on the relationship between local government debt and economic growth. Some studies argue that local government debt promotes economic growth. Adelino et al. (2017) find that municipal debt financing expands fiscal spending and generates significant spillover effects on private consumption and investment during economic downturns, thereby

stimulating regional economic growth. From an infrastructure perspective, [Lin et al. \(2023\)](#) show that local government debt reduces transaction costs and improves the return on capital and investment growth of non-state-owned enterprises, thus exerting a positive effect on economic growth. In contrast, another strand of literature emphasizes the crowding-out and suppressive effects of government debt on growth. [Ahlborn and Schweickert \(2018\)](#), using samples from countries with weak fiscal sustainability, find a significant negative relationship between government debt levels and GDP growth. [Liu and Li \(2021\)](#) further point out that the accumulation of local government debt crowds out liquidity in the real economy, causing the output growth effect of new debt to decline over time.

Other studies suggest that the relationship between government debt and economic growth is nonlinear and characterized by threshold effects. [Minea and Parent \(2012\)](#) find that economic growth performs better when government debt is below 60% of GDP, but slows significantly when the ratio exceeds 90%. [Han and Cheng \(2018\)](#) document an inverted U-shaped relationship between local government debt and economic growth, indicating that once debt surpasses a certain threshold, its effect shifts from promoting to inhibiting growth.

Overall, macro-level studies highlight the “double-edged sword” nature of local government debt, with its growth effects depending critically on debt scale, fund allocation, and fiscal sustainability. However, this line of research largely remains at the aggregate level and provides limited insight into the micro-level channels through which local government debt affects firm behavior, particularly the efficiency of corporate investment decisions.

2.1.2. Microeconomic effects of local government debt

At the micro level, existing studies mainly examine changes in corporate financing behavior and innovation activities under local government debt expansion. Research on corporate financing focuses on financing scale, cost, and structure. [Wang et al. \(2020\)](#) argue that government debt financing intensifies competition for funds between the public and corporate sectors, significantly reducing listed firms' financing scale. [Yu and Kang \(2020\)](#) find that local government debt raises financing costs for local firms by increasing demand in financial markets. [Chang et al. \(2023\)](#) further show that debt expansion crowds out bank lending to non-state-owned enterprises, forcing firms to rely more on equity and internal financing. Regarding corporate innovation, studies by [Xiong and Shen \(2019a, 2019b\)](#) and [Liu et al. \(2020\)](#) demonstrate that local government debt exacerbates financing constraints and crowds out high-risk, high-uncertainty innovation activities, thereby inhibiting firms' innovation investment.

While these micro-level studies provide important evidence on the firm-level effects of local government debt, they mainly focus on intermediate outcomes such as financing constraints and innovation input, rather than directly examining firms' ultimate investment efficiency. Whether local government debt leads firms to “invest incorrectly” or “invest inefficiently” remains an open question.

2.2. Causes of corporate inefficient investment

Corporate inefficient investment refers to deviations of actual investment from the optimal level and typically manifests as overinvestment or underinvestment. Overinvestment occurs when firms invest in projects with negative net present value (NPV), whereas underinvestment arises when firms forgo projects with positive NPV. Both forms of inefficient investment damage firm value and hinder long-term development.

Existing literature explains the causes of inefficient investment mainly through information asymmetry and agency problems. On the one hand, information asymmetry between firms and external investors raises external financing costs, leading firms to abandon profitable projects under financial constraints and resulting in underinvestment. On the other hand, within the agency framework, managerial risk preferences and distorted incentives can also induce inefficient investment. [He and Mao \(2023\)](#) note that managers may avoid high-return projects to reduce risk, causing underinvestment; when free cash flow is abundant, managers may pursue empire building for private benefits, leading to overinvestment.

Further studies show that inefficient investment is influenced by a variety of internal and external factors. Internally, [Biddle et al. \(2009\)](#), [Cheng et al. \(2013\)](#), and [Wang et al. \(2015\)](#) find that high-quality internal controls mitigate agency problems and information asymmetry, thereby reducing inefficient investment. [Habib and Hasan \(2017\)](#) argue that highly capable managers are more prone to overinvestment, while [Wu and Zheng \(2017\)](#) show that managerial overconfidence induces overinvestment when cash flows are sufficient and exacerbates underinvestment when funds are tight. [Panousi and Papanikolaou \(2012\)](#), as well as [Wang and Li \(2022\)](#), find that financing constraints reduce investment in high-risk, positive-NPV projects, worsening inefficient investment.

Externally, [Fang et al. \(2017\)](#) and [Sun et al. \(2019\)](#) suggest that media attention can curb managerial irrational investment through external monitoring and public scrutiny, whereas [Liu et al. \(2021\)](#) find that rising economic policy uncertainty disrupts investment expectations and aggravates inefficient investment.

Overall, the literature on corporate inefficient investment has developed a relatively mature analytical framework, but it focuses mainly on internal governance and external institutional environments, paying insufficient attention to fiscal behavior—particularly local government debt—as an important macro factor. This gap provides room for research from a fiscal perspective.

2.3. The effect of government debt on corporate investment

Research on the impact of government debt on corporate investment mainly examines its effect on the scale and structure of corporate investment.

Regarding investment scale, one stream of literature emphasizes the crowding-out effect of government debt expansion. Public debt can crowd out corporate investment by absorbing credit resources and raising financing costs ([Cavallo & Daude, 2011](#); [Huang et al., 2018](#); [Yu and Kang, 2020](#)). Moreover, under debt pressure, governments may strengthen tax enforcement, increasing firms' tax

burdens and further crowding out investment (Chen et al., 2023; Wu et al., 2022).

Another stream highlights potential crowding-in effects. Arrow and Kurz (1970) argue that debt accumulation stimulates private investment and labor supply by promoting public investment. Aschauer (1989) finds that public investment in infrastructure significantly boosts private investment. In the Chinese context, Fan and Mo (2014) show that local government debt promotes local industrial investment through infrastructure construction and lower industrial land prices. Mao et al. (2022) find that urban investment bonds reduce local investment's dependence on savings and attract capital inflows, thereby expanding corporate investment.

Regarding investment structure, Chang et al. (2024) find that local government debt expansion alters firms' financing structures, increasing financial investment while crowding out real investment. Kong and Xie (2021) show that local government debt affects corporate expectations and regional real estate markets, exacerbating firms' shift from the real to the financial sector. In contrast, Gao et al. (2023) argue that under tight credit conditions, local debt expansion may suppress firms' holdings of financial assets, especially short-term ones.

Although these studies reveal the complex effects of government debt on corporate investment from different perspectives, they largely focus on investment scale and direction, with limited attention to whether investment deviates from the optimal level—that is, investment efficiency. This limitation constrains a comprehensive understanding of the microeconomic consequences of government debt.

2.4. Literature summary and commentary

In summary, existing studies on the economic consequences of local government debt focus mainly on its relationship with economic growth at the macro level and on corporate financing, innovation, and investment scale at the micro level (Colak et al., 2025; Fan et al., 2022; Nie et al., 2023; Zhou et al., 2023). Meanwhile, the literature on corporate inefficient investment primarily examines corporate governance, managerial characteristics, financing constraints, and external institutional environments, with relatively little attention to fiscal behavior—especially local government debt (Chen et al., 2024; Zhang, Pan, & Fan, 2022; Zhu et al., 2024). Against the backdrop of rapidly rising local government debt and widespread inefficient investment among firms in China (Shen and Shi, 2021; Liu et al., 2023), whether and how local government debt affects corporate inefficient investment through multiple channels remains an open question. By examining corporate inefficient investment from the perspective of local government debt expansion, this study provides an important supplement to the existing literature.

3. Theoretical analysis and research hypotheses

Inefficient investment generally refers to firms' investment decisions that deviate from the objective of firm value maximization, and it mainly takes two forms: underinvestment and overinvestment. Underinvestment occurs when firms forgo projects with positive net present value (NPV), resulting in actual investment falling below the optimal level; overinvestment refers to firms investing in projects with negative NPV, causing actual investment to exceed the optimal level (Xing et al., 2020). Both forms of inefficient investment lead to resource misallocation and impair firms' long-term performance.

3.1. Local government debt, fiscal spending multiplier effects, and corporate inefficient investment

According to Keynesian demand management theory, when effective demand is insufficient, governments can stimulate aggregate demand by expanding fiscal expenditure through debt financing, thereby generating a fiscal spending multiplier effect in the short run. A salient feature of local government debt in China is that its funds are primarily allocated to infrastructure construction. The National Audit Office's (2013) audit report on local government debt shows that a large proportion of debt funds is invested in infrastructure sectors such as transportation and municipal engineering, which directly stimulates demand in high-investment industries such as cement and steel (Xiong & Shen, 2019a, 2019b). By increasing demand for capital goods, expanding employment, and raising household income, public investment further stimulates investment activities in both consumer goods and capital goods sectors (Lin et al., 2023).

Existing studies have verified from different perspectives the possibility that local government debt promotes private investment through multiplier effects. Liu et al. (2020) find that when the growth rate of implicit debt remains within a reasonable range, local government debt can stimulate public investment and generate multiplier effects, thereby boosting private investment. Wang and Zhu (2021) show that government investment exhibits a significant “crowding-in effect” on private investment during the transmission process. Liu and Li (2021) argue that when local governments allocate debt funds to public investment, total social investment demand expands in the short term, raising private sector investment.

However, under China's unique political incentive and fiscal systems, local government public investment does not necessarily aim at efficiency maximization. For a long time, local officials' promotion incentives have been closely tied to regional GDP growth. Against this background, local governments tend to favor infrastructure projects with rapid and visible outcomes and large scales, while paying less attention to long-term returns and investment efficiency (Peng et al., 2018). Debt financing softens local governments' budget constraints to some extent, making them more prone to investment impulses, redundant construction, and inefficient expansion. When excessive public investment is transmitted to the private sector through multiplier effects, firms may blindly expand investment under the combined influence of demand expansion and government signals, deviating from the optimal investment level and aggravating inefficient investment. Based on the above analysis, this study proposes the following hypothesis:

Hypothesis 1. (H1): When local government debt is used for public investment, it promotes corporate investment expansion through multiplier effects and exacerbates corporate inefficient investment.

3.2. Local government debt, financialization, and corporate inefficient investment

In recent years, China's economy has shown a tendency toward “shifting from the real to the financial sector,” which at the micro level is manifested by firms allocating more resources to financial assets rather than real investment (Kong & Xie, 2021). The expansion of local government debt may affect firms' financial asset allocation through expectation effects and arbitrage effects.

First, the expectation effect operates mainly through the liquidity reserve function of financial assets. Local government implicit debt is closely associated with the shadow banking system (Chen et al., 2020), whose high leverage and weak regulation lead to the continuous accumulation of debt risks. As local government debt expands, concerns about debt sustainability intensify, increasing uncertainty faced by firms and making future expectations more pessimistic. Under such circumstances, firms are more inclined to hold highly liquid financial assets for precautionary savings to cope with potential shocks (Liu et al., 2021). In addition, financial assets—especially derivatives—possess hedging and risk-diversification functions, enabling firms to offset macroeconomic and fiscal uncertainties through financial investments (Peng & Lin, 2024).

Second, the arbitrage effect operates through the investment opportunity attributes of financial assets. On the one hand, the accumulation of government debt risks may push up yields on government bonds at maturity and, through asset pricing mechanisms, raise returns on other risky assets (Yu & Wang, 2022). On the other hand, land collateral financing and land transfer revenues are important supports for local government debt (Zhang et al., 2019). Debt expansion intensifies local governments' reliance on land finance and may drive up real estate prices through capitalization mechanisms. In this context, both financial and real estate markets provide firms with more attractive financial returns, inducing them to increase the proportion of financial asset holdings.

Excessive corporate financialization can significantly exacerbate inefficient investment through multiple channels. On the one hand, financial asset allocation occupies substantial funds and crowds out resources for real operations and long-term projects, leading to insufficient R&D and fixed asset investment. On the other hand, the high expected returns and short-term nature of financial investments may strengthen managerial speculation motives and risk preferences, inducing overinvestment that deviates from firms' core businesses. Meanwhile, the opacity of financial assets intensifies agency problems, enabling managers to use financialization for short-term profit-seeking or earnings management, further distorting investment decisions. Overall, excessive financialization simultaneously triggers underinvestment and overinvestment through crowding-out, speculative, and agency effects, thereby significantly reducing investment efficiency (Zhang et al., 2018). Based on the above analysis, this study proposes the following hypothesis:

Hypothesis 2. (H2): The expansion of local government debt exacerbates corporate inefficient investment by increasing the degree of corporate financialization.

3.3. Local government debt, tax burden, and corporate inefficient investment

Tax burden is a key factor influencing firms' investment decisions, and local government debt may affect firms' effective tax burdens by altering governments' taxation behavior. The impact of local government debt on tax burden may operate through two opposing mechanisms.

On the one hand, local government debt may reduce firms' effective tax burdens through tax–debt substitution and tax competition effects. According to the Ricardian equivalence perspective, government debt and taxation are substitutes in the long run. Debt financing through local government financing vehicles is characterized by low financing costs, weak supervision, and reliance on land transfer revenues for repayment, which may generate fiscal illusion and reduce governments' tax collection efforts (Hong et al., 2024). Moreover, debt financing alleviates short-term fiscal pressure, making local governments more inclined to stimulate economic activity through tax cuts and relaxed tax enforcement (Liu, 2021). From the perspective of tax competition, debt expansion enlarges local governments' fiscal space, enabling them to lower effective tax rates to compete for capital and production factors across regions (Pu et al., 2022; Wu et al., 2022). Although a lower tax burden increases firms' after-tax cash flows, it does not necessarily improve investment efficiency. Reduced tax constraints and increased disposable funds may strengthen managerial opportunism, leading to resource allocation toward low-efficiency or even negative-NPV projects. Furthermore, excessive tax competition may allow lower tax burdens to offset losses from low-return projects, inducing firms to expand investment scale and exacerbating overinvestment (Pu & Yang, 2020; Shen & Shi, 2021, pp. 85–95).

On the other hand, the expansion of local government debt may also increase governments' tax collection efforts. Tax revenue is a key factor affecting market confidence in local governments' debt repayment capacity (Zhang, Pan, & Fan, 2022). As debt scales expand and repayment pressures rise, local governments may strengthen tax enforcement to increase fiscal revenues and maintain refinancing capacity (Chen et al., 2023; Xiong et al., 2024). Stronger tax enforcement may exert an external governance effect that helps restrain irrational managerial investment behavior; however, excessive tax burdens can crowd out funds available for production and operations, dampen investment incentives, and ultimately lead to underinvestment.

Taken together, the impact of local government debt on corporate inefficient investment through the tax channel is ambiguous. Based on these competing mechanisms, this study proposes the following competing hypotheses:

Hypothesis 3a. (H3a): The expansion of local government debt exacerbates corporate inefficient investment by reducing firms' effective tax burdens and inducing overinvestment.

Hypothesis 3b. (H3b): The expansion of local government debt exacerbates corporate inefficient investment by increasing firms' effective tax burdens and causing underinvestment.

4. Research design

4.1. Model specification

To examine the relationship between local government debt and firm inefficient investment, this study establishes the following empirical model:

$$\text{Inveff}_{i,t} = \alpha_0 + \alpha_1 \text{Debratio}_{c,t} + \alpha_2 X_{i,t} + \pi_i + \rho_t + \varepsilon_{i,t} \quad (1)$$

In equation (1), the subscripts i and t denote firm and year, respectively. The dependent variable $\text{Inveff}_{i,t}$ represents the degree of inefficient investment of firm i in year t ; the key explanatory variable $\text{Debratio}_{c,t}$ measures the local government debt of city c in year t ; $X_{i,t}$ denotes a set of firm-level control variables. π_i and ρ_t are firm and year fixed effects, controlling for time-invariant firm characteristics and common shocks in each year, respectively. $\varepsilon_{i,t}$ is the random error term, with standard errors clustered at the city level. The main focus is on the sign and magnitude of the coefficient α_1 .

To further explore the mechanisms through which local government debt affects firm inefficient investment, following Jiang (2022) on mechanism analysis, we construct the following mediation model:

$$\text{MV}_{i,t} = \gamma_0 + \gamma_1 \text{Debratio}_{c,t} + \gamma_2 X_{i,t} + \pi_i + \rho_t + \varepsilon_{i,t} \quad (2)$$

In model (2), $\text{MV}_{i,t}$ represents the mediating variable. In the regressions, $\text{MV}_{i,t}$ is alternately specified as *Pubinv*, *Finlevel* and *Taxburden* representing the multiplier effect, firm financialization level, and firm tax burden, respectively.

4.2. Variable definitions

4.2.1. Inefficient investment (*Inveff*)

The core dependent variable in this study is inefficient investment (*Inveff*). Following the estimation approach proposed by Richardson (2006), we construct the following regression equation to estimate firms' expected level of capital investment:

$$\begin{aligned} \text{Invest}_{i,t} = & \beta_0 + \beta_1 \text{Growth}_{i,t-1} + \beta_2 \text{Cash}_{i,t-1} + \beta_3 \text{Lever}_{i,t-1} + \beta_4 \text{Return}_{i,t-1} + \beta_5 \text{Age}_{i,t-1} + \beta_6 \text{Size}_{i,t-1} + \beta_7 \text{Invest}_{i,t-1} + \sum \text{Year} \\ & + \sum \text{Indust} + \mu_{i,t} \end{aligned} \quad (3)$$

where i and t denote firms and years, respectively; β represents the estimated parameters; and $\mu_{i,t}$ is the error term. *Invest* denotes the firm's investment level in year t , measured as cash paid for the acquisition and construction of fixed assets, intangible assets, and other long-term assets scaled by total assets. *Growth* represents growth opportunities and is proxied by Tobin's Q , defined as the firm's market value divided by total assets. *Cash* denotes cash holdings, measured as year-end cash divided by total assets. *Lever* represents financial leverage, measured as total liabilities divided by total assets. *Return* denotes stock returns, measured by the annual individual stock return adjusted for cash dividend reinvestment. *Age* represents listing age, calculated as the difference between the current year and the firm's listing year. *Size* denotes firm size and is measured as the natural logarithm of total assets. *Year* and *Indust* denote year and industry dummy variables, respectively, with industry classification based on the China Securities Regulatory Commission (CSRC) industry classification standard.

The absolute value of the residuals obtained from this model is used as a proxy for firms' inefficient investment (*Inveff*). A larger value indicates lower investment efficiency and a higher degree of inefficient investment. When the residual is greater than zero, the firm is classified as exhibiting overinvestment (*Ovinv*); when the residual is less than zero, the firm is classified as exhibiting underinvestment (*Uninv*).

4.2.2. Local government debt ratio (*debratio*)

The core explanatory variable in this study is local government debt. Given inconsistencies in measurement across data sources, this study adopts the ratio of the flow of urban investment bonds issued by local government financing vehicles to GDP as a proxy, referred to as the local government debt ratio (*Debratio*) (Qu et al., 2023; Yang et al., 2023).

4.2.3. Mechanism variables

To examine the mechanisms through which local government debt affects corporate inefficient investment, the following mediating variables are employed:

- (1) Public investment (*Pubinv*): As indicated in the theoretical analysis, local government debt can generate a multiplier effect by stimulating public sector investment. Following Liu and Li (2021), public investment is used to capture the multiplier effect and is measured as the ratio of state-owned fixed asset investment to total fixed asset investment at the prefecture-city level. A higher ratio indicates a greater scale of public investment.

- (2) Corporate financialization (*Finlev*): Following Du et al. (2019), corporate financialization is measured as the ratio of financial assets to total assets. A higher value indicates a higher degree of financialization. Financial assets include trading financial assets, derivative financial assets, loans and advances, available-for-sale financial assets, held-to-maturity investments, and investment properties.
- (3) Corporate tax burden (*Tax*): The corporate tax burden is measured using the firm's effective tax rate.

4.2.4. Control variables

Following the existing literature (Chen et al., 2022; Zheng et al., 2025), this study controls for a set of firm characteristics that may affect inefficient investment:

- (1) Firm size (*Size*): measured as the natural logarithm of total assets;
- (2) Listing age (*Age*): calculated as the observation year minus the firm's listing year plus one, and expressed in natural logarithms;
- (3) Financial leverage ratio (*Finlev*): measured as the sum of non-current liabilities, short-term borrowings, non-current liabilities due within one year, trading financial liabilities, and derivative financial liabilities divided by total liabilities;
- (4) Total asset turnover (*Assetturnover*): measured as operating revenue divided by the ending balance of total assets;
- (5) Revenue growth rate (*Regrowth*): measured as the difference between current-year operating revenue and prior-year operating revenue divided by prior-year operating revenue;
- (6) Management expense ratio (*Manfee*): measured as management expenses divided by main business revenue;
- (7) Ownership concentration (*Top5hhi*): measured using the Herfindahl–Hirschman Index (HHI) of shareholdings of the top five shareholders.

4.3. Sample selection and data sources

This study uses Chinese A-share listed firms from 2008 to 2021 as the research sample. Following existing literature, the data are processed as follows: (1) ST firms are excluded; (2) financial firms are excluded; and (3) observations with missing key variables are removed. Firm-level data are matched with prefecture-level city data to construct a combined dataset. To mitigate the influence of extreme values, all continuous firm-level variables are winsorized at the 1st and 99th percentiles. The final sample consists of 49,656 firm-year observations. Firm financial data are obtained from the CSMAR database, local government debt data are sourced primarily from the Wind database, and regional macroeconomic data are collected from the *China City Statistical Yearbook* and local statistical yearbooks.

Descriptive statistics for the main variables are reported in Table 1.

5. Empirical results analysis

5.1. Baseline regression

We first estimate equation (2) using the full sample to examine the impact of local government debt expansion on firm inefficient investment. The results are reported in Table 2. In column (1), only the local government debt ratio (*Debratio*) is included, showing that local government debt expansion significantly increases the level of firm inefficient investment. Columns (2)–(5) gradually introduce firm-level control variables, and the main conclusion remains robust.

Further, the results in column (5) indicate that firms with higher revenue growth rates and greater ownership concentration exhibit more severe inefficient investment, reflecting that optimistic growth expectations and controlling shareholders' opportunistic motives may induce overinvestment. In contrast, firm age, total asset turnover, and management expense ratio are all significantly negatively correlated with inefficient investment, suggesting that stronger governance experience, higher asset utilization efficiency, and internal

Table 1
Descriptive statistics.

Variables	Definition	(1) N	(2) mean	(3) sd	(4) min	(5) max
Inveff	Inefficient investmen	49,656	0.0191	0.0302	0.0000	0.1900
Size	Firm's total asset	39,726	22.0028	1.3725	18.9884	27.0906
Age	Firm's age	38,680	2.0369	0.9407	0.0000	3.3322
Finlev	Firm's financial debt ratio	49,346	0.3870	0.2569	0.0000	0.9021
Assetturnover	Total asset turnover of firms	46,317	0.7136	0.4830	0.0259	2.7551
Regrowth	Firm's manage fee ratio	46,292	0.1971	0.4538	−0.6268	3.0038
Manfee	Growth rate of firm's revenue	49,280	0.0002	0.0001	0.0001	0.0007
Top5hhi	Firm's ownership concentration	39,052	53.6809	15.7059	0.8109	99.2300
Debratio	Debt ratio of local government	49,656	0.0010	0.0022	0.0000	0.0133
Pubinv	Public investment	46577	0.3000	0.1150	0.0000	0.9817
Financelevel	Firm's financial level	37,797	0.0390	0.0723	0.0000	0.4584
Tax	Firm's tax burden	49,336	0.2897	0.3693	−0.6418	2.3299

Table 2
Baseline regression results.

	(1)	(5)
VARIABLES	Inveff	Inveff
Debratio	0.2911* (0.150)	0.1919* (0.105)
Size		0.0005 (0.000)
Age		−0.0023*** (0.001)
Finlev		0.0030** (0.001)
Assetturnover		−0.0044*** (0.001)
Regrowth		0.0014*** (0.000)
Manfee		−8.3124** (4.212)
Top5hhi		0.0002*** (0.000)
Constant	0.0188*** (0.000)	0.0107 (0.010)
Year	Yes	Yes
Firm	Yes	Yes
N	49,296	37,541
Adj R ²	0.403	0.521

Notes: Robust standard errors in parentheses, ***p < 0.01, **p < 0.05, *p < 0.1, Year and Firm represent control fixed effect of year and firm respectively.

control mechanisms help restrain firm inefficient investment.

5.2. Robustness checks

5.2.1. Replacing the core explanatory variable

First, we replace the local government debt ratio (*Debratio*) with *Debtfr* (the flow of local government financing platform bonds as a share of fiscal revenue) in the regression. The results are reported in column (1) of Table 3. The results indicate that local government debt expansion still significantly increases firm inefficient investment, suggesting that this effect is not driven by a single measure of debt but reflects the general impact of local government financing behavior on firm investment decisions.

Second, we replace *Debratio* with the logarithm of the city investment bonds (*Lndebt*) and its squared term (*Lndebt2*), as shown in column (2) of Table 3. The results reveal a significant “U-shaped” relationship between local government debt size and firm inefficient investment. This suggests that at relatively low debt levels, government borrowing may suppress firm inefficient investment by improving the financing environment and stabilizing economic expectations. However, once debt exceeds the critical threshold

Table 3
Robustness check results.

	(1)	(2)	(3)	(4)
VARIABLES	Inveff	Inveff	Inveff	Inveff
Debtfr	0.0140* (0.007)			
Lndebt		−0.0008*** (0.000)		
Lndebt2		0.0001*** (0.000)		
Debratio			0.3810*** (0.142)	0.1714* (0.095)
Constant	0.0105 (0.010)	0.0113 (0.011)	0.0150 (0.012)	0.0187** (0.009)
Controls	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes
Firm	Yes	Yes	Yes	Yes
Industry	No	No	No	Yes
N	37,541	36,968	29,635	37,541
Adj R ²	0.521	0.521	0.525	0.525

Notes: Robust standard errors in parentheses, ***p < 0.01, **p < 0.05, *p < 0.1, Year, Firm and Industry represent control fixed effect of year, firm and industry respectively.

(2.8748), the resource misallocation and crowding-out effects dominate, significantly exacerbating firm inefficient investment. Given that approximately 75% of the sample exceeds this threshold, this finding has strong practical relevance for the overall sample.

5.2.2. Excluding the impact of COVID-19

The COVID-19 pandemic caused demand contractions and supply constraints, potentially affecting investment behavior through changes in firm expectations and cash flows. To avoid bias from the pandemic shock, we exclude observations from 2020 onward. The results, reported in column (3) of Table 3, show that even after removing the pandemic period, local government debt expansion still significantly increases firm inefficient investment at the 1% level. This indicates that the previous conclusions are not driven by extreme macro shocks and are robust.

5.2.3. Changing the regression model

On the basis of the baseline model with firm and year fixed effects, we further introduce industry fixed effects to eliminate the influence of industry-specific cycles and investment characteristics. The results are reported in column (4) of Table 3. The results show that the promoting effect of local government debt expansion on firm inefficient investment remains significant, suggesting that this effect is not driven by specific industries but has broader economic implications, further reinforcing the robustness of the findings.

5.3. Endogeneity test

Given that omitted variables, measurement errors, and potential reverse causality may lead to endogeneity bias in the regression results, this study employs an instrumental variable (IV) approach for testing. In China, “developing through land” has long been an important strategy for local governments to increase fiscal revenue and stimulate economic growth (Liu et al., 2020a; Zhu et al., 2022). Local governments obtain revenue through strategic land sales, and land sale prices and revenues are often constrained by urban topography: land with gentle slopes is more suitable for population concentration and urban construction, yielding higher per-unit land sale prices; conversely, steep land is harder to develop, more costly, and commands lower sale prices, thereby limiting local governments' ability to generate sufficient fiscal revenue (Liu et al., 2023; Yang et al., 2023). Under this context, regions with less favorable terrain are more likely to rely on borrowing to fill fiscal gaps and promote economic growth, making local government debt levels correlated with land slope (Fan et al., 2022; Zheng et al., 2025). At the same time, land slope, as a natural geographic feature, does not directly affect firms' investment decisions or inefficient investment, satisfying the exogeneity requirement of an instrumental variable.

To accommodate panel data regression, we construct an instrument by interacting the city-level average land slope with a time variable and taking the natural logarithm:

$$IV = \ln(\text{slopet}) = \ln(\text{Slope} \times t), \text{ where } t = \text{year} - 2007$$

This specification not only captures the long-term constraint of topography on local government fiscal capacity but also accounts for potential policy and economic dynamic effects over time. As we can see in Table 4, In the regression, $\ln(\text{slopet})$ is used as an instrument for the local government debt ratio (*Debratio*), and estimation is performed using the IV-2SLS method. The results show that in the first-stage regression, land slope has a significant positive effect on local government debt ratio, indicating a strong correlation between the instrument and the core explanatory variable. The second-stage regression results indicate that, after addressing endogeneity, local government debt ratio still significantly increases firm inefficient investment.

This further confirms the robustness of the conclusion that local government debt expansion exacerbates firm inefficient investment through capital misallocation and crowding-out effects, while highlighting the institutional role of fiscal behavior in shaping corporate investment decisions.

Table 4
Endogeneity test results.

Variables	first stage	second stage
	Debratio	Inveff
Lnslopet	0.0003*** (0.000)	
Debratio		4.5068*** (0.508)
Constant	−0.0003 (0.000)	0.0893*** (0.003)
Controls		
Year	Yes	Yes
Firm	Yes	Yes
N	38,107	38,107
Adj R ²	0.031	

Notes: Robust standard errors in parentheses, ***p < 0.01, **p < 0.05, *p < 0.1, Year, Firm and Industry represent control fixed effect of year, firm and industry respectively.

5.4. Mechanism test

Firm inefficient investment mainly manifests in two forms: overinvestment and underinvestment. The baseline regression results show that local government debt expansion significantly increases the level of firm inefficient investment. To further distinguish its specific manifestations, Columns (1) and (2) of Table 5 report regressions using overinvestment and underinvestment as dependent variables, respectively. The results indicate that local government debt significantly exacerbates overinvestment while, to some extent, alleviating underinvestment. Moreover, the promoting effect on overinvestment is markedly stronger than the mitigating effect on underinvestment, thereby overall raising the level of firm inefficient investment.

On this basis, we further examine the mechanisms through which local government debt affects firm inefficient investment via three channels: public investment, firm financialization, and corporate tax burden. First, Column (3) of Table 5 shows that local government debt significantly increases public investment. Existing studies suggest that public investment can stimulate private investment through a multiplier effect (Liu et al., 2020b; Wang & Zhu, 2021; Liu & Li, 2021). Under the constraints of officials' promotion incentives, local governments often have a tendency to overinvest by expanding project scale to boost GDP growth (Wu & Wang, 2018; Peng & Wu, 2023, pp. 81–94). Local government debt provides funding support for such investment expansion, which in turn induces overinvestment by firms via the multiplier effect, significantly aggravating inefficient investment.

Second, Column (4) of Table 5 shows that local government debt significantly increases firm financialization. Specifically, a 1% increase in local government debt ratio raises the level of firm financialization by approximately 0.9185%. This may be because debt expansion heightens firms' concerns over macroeconomic conditions and fiscal sustainability while increasing the relative returns of financial assets such as real estate, thereby reinforcing firms' precautionary and arbitrage-driven allocation to financial assets.

Third, Column (5) of Table 5 shows that local government debt significantly reduces the corporate tax burden, with a 1% increase in debt ratio lowering firm tax burden by about 5.459%. This is likely because debt expansion partially relieves local fiscal pressure, generating a substitution effect for tax revenue and providing fiscal space for local governments to attract factor inflows through tax competition. A lower tax burden may, on one hand, increase free cash flow and induce managerial opportunism, thereby expanding overinvestment; on the other hand, it may reduce financing constraints and mitigate underinvestment. Column (6) of Table 5 further shows that corporate tax burden is significantly negatively associated with overinvestment, indicating that local government debt, by lowering taxes, can stimulate overinvestment.

In summary, the mechanism analysis indicates that: local government debt significantly exacerbates overinvestment by increasing public investment and leveraging the investment multiplier; it partially restrains overinvestment and reduces inefficient investment by raising firm financialization; and it simultaneously expands overinvestment and mitigates underinvestment by lowering corporate tax burden, with a stronger effect on overinvestment, ultimately raising overall firm inefficient investment. These results provide strong empirical support for Hypotheses 1, 2, and 3 b.

5.5. Heterogeneity analysis

5.5.1. Firm-level heterogeneity analysis

First, based on financing constraints, sample firms are divided into two groups according to the median absolute value of the SA index: firms with weaker financing constraints and firms with stronger financing constraints. Group regressions are then conducted separately. Columns (1) and (2) of Table 6 show that local government debt expansion significantly promotes inefficient investment for firms with weaker financing constraints, while the inhibitory effect for firms with stronger financing constraints exists but is not significant. This indicates that firms with weaker financing constraints can more easily access funds and thus benefit from the improved financing environment brought by local government debt expansion, but they are also more prone to irrational or excessive investment. In contrast, firms with stronger financing constraints are limited in their access to funds, so even when local government debt expands, they are less able to increase investment, resulting in an insignificant effect on inefficient investment.

Second, based on internal control levels, firms are divided into groups with weaker or stronger internal control using the median management expense ratio. Columns (3) and (4) of Table 6 indicate that local government debt expansion significantly increases inefficient investment for firms with weaker internal control, while the effect on firms with stronger internal control is not significant.

Table 5
Mechanisms test.

VARIABLES	(1) Ovinv	(2) Uninv	(3) Pubinv	(4) Financelevel	(5) Tax	(6) Ovinv
Debratio	0.3807*** (0.110)	−0.1888*** (0.065)	19.0689*** (4.925)	0.9185* (0.539)	−5.4590*** (1.841)	−0.0009*** (0.0003)
Constant	0.0367*** (0.013)	−0.0260*** (0.008)	0.3921*** (0.069)	0.1755*** (0.035)	0.2158*** (0.133)	0.0401*** (0.0105)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes	Yes
Firm	Yes	Yes	Yes	Yes	Yes	Yes
N	37,541	37,541	12,287	36,881	38821	38821
Adj R ²	0.443	0.659	0.718	0.581	0.265	0.4391

Note: Robust standard errors in parentheses, ***p < 0.01, **p < 0.05, *p < 0.1.

Table 6

Firm-level heterogeneity analysis.

	Financial constraints		Internal Control		Firm's size	
	weak (1)	Strong (2)	Weak (3)	Strong (4)	Big (5)	Small (6)
	Inveff	Inveff	Inveff	Inveff	Inveff	Inveff
Debratio	0.7240*** (0.271)	−0.1996 (0.184)	0.2417* (0.136)	0.0225 (0.211)	0.2311* (0.126)	0.0206 (0.225)
Constant	−0.0065 (0.021)	0.0007 (0.019)	0.0148 (0.019)	0.0286 (0.018)	0.0028 (0.019)	0.0224 (0.022)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes	Yes
Firm	Yes	Yes	Yes	Yes	Yes	Yes
N	18,200	18,839	18,615	18,281	19,196	17,918
Adj R ²	0.532	0.635	0.620	0.494	0.664	0.487

Note: Robust standard errors in parentheses, ***p < 0.01, **p < 0.05, *p < 0.1.

This suggests that firms' internal governance can partially buffer the impact of external fiscal changes on investment behavior. Firms with stronger internal control typically have more robust investment decision-making mechanisms and risk management systems, allowing them to maintain relatively cautious investment strategies even in a relaxed external financing environment. In contrast, firms with weaker internal control are more easily influenced by policy and financing environment changes, increasing the likelihood of inefficient investment.

Finally, based on firm size, firms are divided into large and small firms using the median total assets. Columns (5) and (6) of Table 6 show that local government debt expansion significantly increases inefficient investment for large firms, whereas the effect on small firms is not significant. This may be because large firms generally have stronger financing capacity and resource allocation ability, making it easier for them to obtain credit support from local government debt expansion and invest more, but they are also more prone to overinvestment or resource misallocation. In contrast, small firms face capital constraints, so external debt expansion has limited impact on their investment behavior, resulting in insignificant changes in inefficient investment.

In summary, firm characteristics such as financing constraints, internal control level, and firm size significantly moderate the impact of local government debt on firm inefficient investment. This indicates that the mechanisms through which local government debt expansion affects firm investment behavior are heterogeneous. These heterogeneity analyses reveal that debt expansion generates different investment responses across firms via channels such as fund accessibility, governance constraints, and scale effects, providing a deeper perspective for understanding the economic consequences of local government debt on firm investment efficiency.

5.5.2. Regional-level heterogeneity analysis

This study examines the heterogeneous effects of local government debt on firm inefficient investment from two regional dimensions: fiscal pressure and economic development level.

First, from the fiscal pressure perspective, fiscal pressure is measured as the ratio of general budgetary revenue to expenditure for each region. Using data from Chinese cities between 2011 and 2021, cities are divided into two groups based on the median: those with lower fiscal pressure and those with higher fiscal pressure. After matching with firm-level data, group regressions are conducted. Columns (1) and (2) of Table 7 show that in cities with lower fiscal pressure, local government debt expansion significantly increases firm inefficient investment, whereas in cities with higher fiscal pressure, this effect is not significant. This difference may arise because cities with lower fiscal pressure have ample fiscal resources, allowing local government debt to be more heavily invested in infrastructure, public services, and local economic development projects. This improves the investment environment and provides firms with more financing channels, making it easier for them to expand investment and generate inefficient investment. In contrast, cities

Table 7

Regional-level heterogeneity analysis.

VARIABLES	Fiscal pressure		Economic Development Level	
	Lower (1)	Higher (2)	Higher (3)	Lower (4)
	Inveff	Inveff	Inveff	Inveff
Debratio	0.3271*** (0.082)	−0.0294 (0.130)	0.3644*** (0.119)	−0.1329 (0.203)
Constant	0.0212 (0.013)	0.0050 (0.017)	0.0173 (0.021)	−0.0042 (0.014)
Controls	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes
Firm	Yes	Yes	Yes	Yes
N	18,523	18,657	19,006	17,572
Adj R ²	0.540	0.564	0.556	0.536

Note: Robust standard errors in parentheses, ***p < 0.01, **p < 0.05, *p < 0.1.

with higher fiscal pressure have limited fiscal space, and debt expansion is less likely to effectively transmit to firms, resulting in a weaker response in inefficient investment.

Second, from the perspective of regional economic development, cities are divided into higher-GDP and lower-GDP groups based on the median city GDP. Columns (3) and (4) of Table 7 show that local government debt expansion significantly promotes inefficient investment in firms located in higher-GDP cities, while the effect is insignificant in lower-GDP cities. This may be because regions with stronger economic foundations have firms with more developed financing systems and greater capacity to absorb and utilize resources. These firms can take full advantage of the improved funding environment brought by local government debt expansion for expansionary investment, even if part of it is inefficient. In contrast, firms in less developed regions are constrained by market size, financing capacity, and technological capabilities, so debt expansion cannot effectively translate into investment behavior, and the effect on inefficient investment is limited.

In summary, regional fiscal space and economic development level significantly moderate the impact of local government debt on firm inefficient investment. These results reveal the mechanisms through which government debt influences firm behavior—by improving the investment environment, facilitating financing, and enabling policy transmission—highlighting the importance of regional heterogeneity in understanding the economic consequences of debt expansion.

6. Conclusion

Based on matched data of Chinese A-share listed companies and prefecture-level macroeconomic indicators from 2008 to 2021, this study systematically examines the impact of local government debt on firm inefficient investment and its underlying mechanisms. The results show that local government debt expansion significantly increases the level of firm inefficient investment, primarily manifested as a marked intensification of overinvestment. Mechanism analysis reveals three main channels: (1) Local government debt expansion significantly boosts public investment, and through the investment multiplier effect, the government's excessive investment is transmitted to firms, thereby exacerbating firm inefficient investment. (2) Local government debt expansion increases firm financialization, which to some extent restrains overinvestment and alleviates inefficient investment. (3) Local government debt expansion reduces corporate tax burdens, increasing firms' free cash flow; however, the effect on promoting overinvestment is stronger than the effect on mitigating underinvestment, ultimately raising the overall level of inefficient investment. Further heterogeneity analysis indicates that the effects of local government debt on firm inefficient investment are mainly concentrated in firms with weaker financing constraints, weaker internal controls, and larger size, as well as in regions with lower fiscal pressure and higher economic development levels.

From the perspective of micro-level firm investment behavior, this study reveals the economic consequences of local government debt expansion, extends research on how government debt influences corporate behavior, and enriches the literature on the determinants of firm inefficient investment. It provides new empirical evidence for understanding the interaction between local governments and firms. The findings carry important policy implications for strengthening local government debt regulation, alleviating firm inefficient investment, and promoting high-quality economic development:

First, local government debt expansion transmits government overinvestment to firms by increasing public investment and leveraging the multiplier effect, driven by officials' promotion incentives linked to regional GDP. Therefore, it is necessary to further improve the official evaluation system, weaken the "GDP-only" approach, strengthen assessments of investment quality, fiscal sustainability, and debt risk, and enhance transparency in the use and allocation of local government debt to improve public investment efficiency.

Second, local government debt expansion may induce firm financialization. Although empirical results show that financialization can partially restrain overinvestment and alleviate inefficient investment, from the perspective of expectation effects and arbitrage motives, the "shift from real to virtual" reflects declining expected returns in the real economy and may accumulate systemic financial risks over the long term, threatening economic stability. Thus, the allocation between government financing and social capital should be optimized to guide financial resources toward the real economy and prevent financialization risks arising from local debt expansion.

Third, local government debt expansion can partially alleviate underinvestment by reducing corporate tax burdens but may also trigger managerial opportunism, exacerbate overinvestment, and lead to resource misallocation and excess capacity. Therefore, when implementing tax incentives and tax competition policies, local governments should adopt a moderate approach, strengthen coordination between tax administration and corporate governance, and enhance supervision of managerial behavior to avoid sacrificing investment efficiency for short-term factor inflows.

Author statement

There is no interests conflict in this article.

Data availability

The data that has been used is confidential.

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